Digital Image Processing

Course Project Instructions and Topic List | 2026

Topic submission: Sunday, 18 October 2026 • Team size: 1 or 2 students

Read the course instructions below before selecting a project. The 62 topics at the end are reference ideas. You may adapt a listed topic or propose your own, provided the approved project contains substantive Digital Image Processing (DIP) work.

1. Topic selection and registration

- Work individually or in a team of two. Teams of three or four are not permitted for this DIP course project.
- Select a topic from the supplied project bank or propose an original topic. Use the project ID if you select a listed topic; otherwise write “Own Topic”.
- Submit the topic through the announced online form by Sunday, 18 October 2026.
- In a two-student team, each student must submit the form separately. Both forms must give the same title or project ID and identify the partner by name and roll number.
- Topics may be returned for revision if the proposed work has insufficient DIP content or substantially overlaps another team’s topic.

2. Project requirements

Every approved project must process image, video, document-image or remote-sensing data and demonstrate the contribution of image processing. A general software application, text-only AI system or one-call model demonstration is not sufficient.

- Define the input images or video, the problem to be solved and the source of the dataset.
- Implement at least one meaningful DIP stage, such as enhancement, histogram operations, filtering, denoising, colour processing, edge or feature extraction, segmentation, morphology, region analysis or change detection.
- Use AI/CV methods when they help solve the problem. These may include classical ML, CNNs, object detection, segmentation networks, ViTs, image embeddings or multimodal models.
- Compare a baseline with an improved or alternative approach. Show both intermediate images and final outputs.
- Report suitable metrics and explain representative success and failure cases.
- Submit reproducible code in Python, MATLAB or another appropriate environment, together with instructions to run it.

An API, cloud platform, database or dashboard is not compulsory unless it is necessary for the selected DIP project. The image-processing and evaluation work must remain the central part of the submission.

3. Details required in the online topic form

- Student name and roll number; individual or two-student team; partner name and roll number, if applicable.
- Project title and either the matching HSR-AIML project ID or “Own Topic”.
- Problem statement (3–5 lines) describing the problem and proposed result.
- Three to five specific DIP or vision concepts that the team will implement.
- Input/data source, expected output and planned method of comparison or evaluation.

4. Work to show in the final demonstration

A suitable sequence is: input images/video → preprocessing → image analysis or features → model (if used) → output → evaluation.

- Working code with a README or equivalent instructions and a description of the dataset.
- A block diagram of the processing pipeline, showing where DIP operations are used.
- Screenshots or figures showing original inputs, intermediate processing stages and final results.
- A baseline-versus-improved comparison with quantitative results.
- Brief discussion of errors, limitations and failure cases; final report and demonstration/presentation.
- For a two-student project, both members must understand the full pipeline and explain their respective contributions.

Possible measures include accuracy, precision, recall, F1, IoU and Dice for detection/segmentation; PSNR or SSIM for image restoration; or a well-justified task-specific score. Execution time and memory use may also be reported where relevant.

5. Using the 62-project reference list

The company listed beside each topic is only an industry reference. It does not indicate sponsorship, an internship, approved access to company data or an official company assignment. Use public datasets, data collected with appropriate consent or clearly documented synthetic data.

Fourteen topics in the bank already mention image or vision work. Their suitable DIP directions are indicated below and in their project briefs. The other topics were written mainly for software, data, financial, IoT, NLP or agentic AI projects. They may be proposed for this course only with a specific and technically justified DIP/vision component.

The original project bank was intended for larger B.Tech project groups. For this DIP course, the team size and deliverables in Sections 1–4 take precedence. Students are not required to reproduce the full cloud, microservice or MLOps scope of the original advanced projects.

6. Projects with an existing image/vision connection

The following 14 entries are the clearest starting points for a DIP project. They still require a defined method, dataset, comparison and measurable outcome.

004 Insurance Document Intelligence for Automated Policy and Claim Intake — Document enhancement, grayscale/contrast, thresholding, morphology, region/character segmentation, OCR accuracy comparison.

013 AI-Assisted 3D Commerce Scene Builder from Product Images and Text — Image preprocessing/segmentation, feature or ViT/VLM embeddings, product-image retrieval/recognition.

014 WebAR Asset Optimization and Delivery Pipeline — Image resizing/interpolation, texture quality, resolution/compression-quality trade-off, visual quality metrics.

016 Computer-Vision Exercise Form Feedback Prototype — Frame preprocessing, foreground/pose extraction, features/keypoints, CNN/pose model, temporal analysis.

029 Multimodal eKYC Identity Verification with Face, Document and Liveness Signals — Face/document enhancement, segmentation, feature extraction, CNN/ViT, image-quality and verification metrics.

033 SAR–Optical Multi-Sensor Change Detection for Disaster Intelligence — Multiband/FCC processing, registration/preprocessing, threshold/change maps, segmentation, CNN/ViT fusion.

034 Air-Gapped Geospatial ML Inference Platform for Multi-Sensor Imagery — Raster tiling, normalization, patch processing, segmentation/detection, stitching and accuracy/speed analysis.

042 Document Intelligence for Application Readiness and Missing-Requirement Detection — Document image preprocessing, contrast/thresholding, morphology, OCR and field extraction.

046 Multimodal Concept Testing for CPG Product Innovation — Colour/texture/shape preprocessing, image features, CNN/ViT/VLM embeddings, visual similarity/retrieval.

049 Camera-Only Road Scene Understanding for Indian Driving Conditions — Edges/Hough vs learnt lane detection, object detection, semantic segmentation, CNN/ViT comparison.

050 Continual Learning Benchmark for Autonomous Driving Domain Shift — Brightness/contrast/weather normalization, augmentation, image-quality/domain analysis, CNN/ViT robustness.

055 Generative Fashion Product Image Studio with Identity and Garment Consistency Checks — Segmentation/masks, resizing/colour consistency, inpainting, perceptual/similarity evaluation.

056 Automated E-commerce Image Quality and Compliance Inspector — Contrast/sharpness/noise measures, segmentation, morphology, object/background analysis, quality classification.

057 Creative Performance Prediction and Automated Variant Ranking for Ads — Image feature/embedding extraction, colour/layout/quality analysis, CNN/ViT/VLM features linked to prediction.

7. Reference topics — complete list

Use the ID below when registering a topic. A project marked “Adapt” requires an additional substantial image-processing component before it can be considered for this course.

ID	Project title	Company	DIP fit
001	Agentic Data Quality Incident Detection and Root-Cause Assistant	Acceldata	Adapt
002	Distributed Data-Pipeline Cost and Performance Optimizer	Acceldata	Adapt
003	Explainable Motor Insurance Claim Triage and Fraud-Risk Scoring Platform	ACKO	Adapt
004	Insurance Document Intelligence for Automated Policy and Claim Intake	ACKO	Direct
005	B2B Supply-Chain Credit Risk Early-Warning Engine	Actyv.ai	Adapt
006	Embedded-Finance Invoice Anomaly and Duplicate Detection Service	Actyv.ai	Adapt
007	Generative Persona Simulator for Customer-Experience Experimentation	Angoor AI	Adapt
008	LLM Evaluation Harness for Brand-Safe Customer Experience Agents	Angoor AI	Adapt

009	Customer Account Health Prediction and Churn-Prevention Copilot	Appeq	Adapt
010	Multi-Agent Customer Success Workflow Orchestrator	Appeq	Adapt
011	Sensitive Data Discovery and Classification Scanner for Cloud Datastores	Aurva	Adapt
012	Data Access Anomaly Detection and Insider-Risk Analytics Platform	Aurva	Adapt
013	AI-Assisted 3D Commerce Scene Builder from Product Images and Text	Ctruh	Vision link
014	WebAR Asset Optimization and Delivery Pipeline	Ctruh	Related
015	Personalized Workout Recommendation and Adherence Prediction System	Cult.fit	Adapt
016	Computer-Vision Exercise Form Feedback Prototype	Cult.fit	Direct
017	IoT Water-Purifier Predictive Maintenance and Filter-Life Forecasting	DrinkPrime	Adapt
018	Smart Service Dispatch Prioritization for Subscription IoT Fleets	DrinkPrime	Adapt
019	Omnichannel Inventory Demand Forecasting and Stockout Prevention Engine	EasyEcom	Adapt
020	Real-Time Order Exception Detection Across E-commerce Channels	EasyEcom	Adapt
021	EV Charging Demand Forecasting and Station Congestion Prediction	ElectricPe	Adapt
022	EV Charger Recommendation and Route-Aware Availability Service	ElectricPe	Adapt
023	B2B Account Intent Scoring and Next-Best-Account Ranking Engine	Factors.ai	Adapt
024	Multi-Touch Marketing Attribution with Causal and Sequence Models	Factors.ai	Adapt
025	Real-Time UPI Transaction Risk Scoring for Youth-Focused Payments	Fam	Adapt
026	Financial Habit Coach with Guardrailed Personalization	Fam	Adapt
027	Real-Time Customer 360 Identity Resolution and Event Unification Pipeline	FirstHive	Adapt
028	RAG-Based Customer Intelligence Copilot for CDP Users	FirstHive	Adapt
029	Multimodal eKYC Identity Verification with Face, Document and Liveness Signals	FRS Labs	Direct
030	Graph-Based Mule Account and Transaction Fraud Detection	FRS Labs	Adapt
031	Intelligent Multi-Channel Notification Routing and Failover Engine	Fyno	Adapt
032	Notification Infrastructure Observability and SLO Anomaly Detection	Fyno	Adapt
033	SAR–Optical Multi-Sensor Change Detection for Disaster Intelligence	GalaxEye Space	Direct
034	Air-Gapped Geospatial ML Inference Platform for Multi-Sensor Imagery	GalaxEye Space	Direct
035	EV Battery Remaining Useful Life Prediction from Telematics	Intellicar	Adapt
036	Fleet Driving Anomaly and Route Deviation Detection Platform	Intellicar	Adapt
037	Personal Finance Transaction Categorization and Cash-Flow Forecasting Engine	Jupiter	Adapt
038	Real-Time Payment Anomaly Detection with Explainable Risk Scores	Jupiter	Adapt
039	Offline-First Merchant Ledger Sync and Conflict Resolution Service	Khatabook	Adapt
040	AI-Assisted Test Automation Platform for Merchant Payment APIs	Khatabook	Adapt
041	Study-Abroad Program Matching and Explainable University Ranking	Leap Scholar	Adapt
042	Document Intelligence for Application Readiness and Missing-Requirement Detection	Leap Scholar	Direct
043	Restaurant Delivery Reconciliation and Revenue Leakage Detection Engine	Loop	Adapt
044	Restaurant Channel Demand Forecasting and Promotion Impact Analytics	Loop	Adapt
045	AI Consumer Insight Mining from Reviews, Social Text and Survey Responses	Machine Vantage	Adapt
046	Multimodal Concept Testing for CPG Product Innovation	Machine Vantage	Vision link
047	Enterprise Sensitive-Data Lineage and Exposure Risk Graph	Matters.ai	Adapt
048	LLM Guardrail for Preventing Sensitive-Data Leakage in Enterprise Assistants	Matters.ai	Adapt
049	Camera-Only Road Scene Understanding for Indian Driving Conditions	Minus Zero	Direct
050	Continual Learning Benchmark for Autonomous Driving Domain Shift	Minus Zero	Vision link
051	Employee Shuttle ETA Prediction and Dynamic Route Optimization	MoveInSync	Adapt
052	Real-Time Fleet Safety and Route-Deviation Monitoring Platform	MoveInSync	Adapt
053	Visitor Access Risk Scoring and Anomaly Detection for Gated Communities	MyGate	Adapt
054	Resident Service Ticket Intelligence and SLA Prediction Platform	MyGate	Adapt
055	Generative Fashion Product Image Studio with Identity and Garment Consistency Checks	Neuropixel.ai	Vision link
056	Automated E-commerce Image Quality and Compliance Inspector	Neuropixel.ai	Direct
057	Creative Performance Prediction and Automated Variant Ranking for Ads	Rocketium	Vision link
058	LLM-Agent Workflow for Brand-Safe Creative Adaptation Across Channels	Rocketium	Adapt
059	Contract Clause Extraction, Risk Classification and Evidence-Grounded Review Assistant	SpotDraft	Adapt
060	Contract Obligation Tracking and Workflow Automation Platform	SpotDraft	Adapt
061	GenAI-Assisted Enterprise Support Portal with Java REST Services	Muraai Information Technologies	Adapt

062	API Integration Testing and Data Reconciliation Suite for Enterprise Systems	Muraai Information Technologies	Adapt

8. Project descriptions

Each brief gives the data, proposed technical work and measures taken from the supplied project bank. Tools are suggestions, not compulsory software requirements. For a DIP course submission, apply the instructions in Sections 1–5.

Common implementation tools: Python, OpenCV, PyTorch, scikit-learn, SQL, Git, FastAPI and Docker, as relevant. Geospatial projects may require rasterio or PostGIS; large-scale data projects may use Spark/Kafka; web applications may use Java or TypeScript. Choose only the tools needed for the approved project.

HSR-AIML-001 Agentic Data Quality Incident Detection and Root-Cause Assistant

Industry reference: Acceldata | Data/AI Infrastructure, DevTools
Data: telemetry, pipeline metadata, data-quality rules and incident history.
Technical work: anomaly detection + causal/rule-assisted RCA + RAG over runbooks.
Evaluation: F1/PR-AUC for incidents; mean time to detect; groundedness; latency.
Relevant tools: Python/Java/Scala; Spark; Kafka; Trino; SQL; FastAPI; Kubernetes; Docker; MLflow; AWS/GCP.

HSR-AIML-002 Distributed Data-Pipeline Cost and Performance Optimizer

Industry reference: Acceldata | Data/AI Infrastructure, DevTools
Data: Spark/Kafka job metrics, cluster utilization and cost data.
Technical work: forecasting + optimization/recommendation for resource configuration.
Evaluation: MAE/MAPE for cost; throughput; runtime reduction; cost reduction.
Relevant tools: Python/Java/Scala; Spark; Kafka; Trino; SQL; FastAPI; Kubernetes; Docker; MLflow; AWS/GCP.

HSR-AIML-003 Explainable Motor Insurance Claim Triage and Fraud-Risk Scoring Platform

Industry reference: ACKO | InsurTech, FinTech
Data: claims tables, policy attributes, repair estimates and event metadata.
Technical work: gradient boosting/graph features for fraud and severity scoring with SHAP explanations.
Evaluation: PR-AUC, recall@fixed precision, calibration, triage latency.

HSR-AIML-004 Insurance Document Intelligence for Automated Policy and Claim Intake

Industry reference: ACKO | InsurTech, FinTech
Data: scanned claim forms, policy PDFs, invoices and IDs.
Technical work: OCR + entity extraction + document classification + evidence-grounded LLM summarization.
Evaluation: Entity F1, document accuracy, extraction accuracy, groundedness.
DIP direction: Document enhancement, grayscale/contrast, thresholding, morphology, region/character segmentation, OCR accuracy comparison.

HSR-AIML-005 B2B Supply-Chain Credit Risk Early-Warning Engine

Industry reference: Actyv.ai | FinTech, B2B Commerce, Supply Chain SaaS
Data: invoice, repayment, order and buyer-supplier transaction histories.
Technical work: time-series/gradient boosting risk model with explainable alerts.
Evaluation: ROC-AUC, PR-AUC, Brier score, recall for delinquency.

HSR-AIML-006 Embedded-Finance Invoice Anomaly and Duplicate Detection Service

Industry reference: Actyv.ai | FinTech, B2B Commerce, Supply Chain SaaS
Data: invoices, purchase orders, payment events and supplier master data.
Technical work: entity matching + anomaly detection + duplicate classification.
Evaluation: Precision/recall/F1, false-positive rate, processing throughput.

HSR-AIML-007 Generative Persona Simulator for Customer-Experience Experimentation

Industry reference: Angoor AI | Generative AI, Customer Experience
Data: product reviews, survey text, behavioral segments and synthetic profiles.
Technical work: LLM-based persona generation with retrieval grounding and consistency checks.
Evaluation: Persona consistency, groundedness, diversity, task success.

HSR-AIML-008 LLM Evaluation Harness for Brand-Safe Customer Experience Agents

Industry reference: Angoor AI | Generative AI, Customer Experience
Data: prompt-response logs, policy documents and test suites.
Technical work: automated LLM-as-judge plus deterministic safety/quality metrics.
Evaluation: Hallucination rate, policy violation rate, groundedness, latency/cost.

HSR-AIML-009 Customer Account Health Prediction and Churn-Prevention Copilot

Industry reference: Appeq | AI SaaS, Customer Success
Data: CRM events, support tickets, product usage and renewal outcomes.
Technical work: churn model + RAG copilot generating evidence-backed next-best actions.
Evaluation: ROC-AUC, recall@k, NDCG, groundedness, action acceptance.

HSR-AIML-010 Multi-Agent Customer Success Workflow Orchestrator

Industry reference: Appeq | AI SaaS, Customer Success
Data: CRM/account events, playbooks, tickets and knowledge base.
Technical work: planner-executor agents with tools, guardrails and human approval.
Evaluation: Task success, tool error rate, latency, cost, human override rate.

HSR-AIML-011 Sensitive Data Discovery and Classification Scanner for Cloud Datastores

Industry reference: Aurva | Cybersecurity, Data Security, DevTools
Data: synthetic database schemas, sample files, logs and labeled PII snippets.
Technical work: rules + NER/classifier for PII/PCI/secret discovery and risk classification.
Evaluation: Entity F1, precision, recall, scan throughput.

HSR-AIML-012 Data Access Anomaly Detection and Insider-Risk Analytics Platform

Industry reference: Aurva | Cybersecurity, Data Security, DevTools
Data: synthetic access logs, user-role metadata and query histories.
Technical work: unsupervised/semi-supervised anomaly detection with explainable risk scores.
Evaluation: PR-AUC, precision@k, false-positive rate, detection latency.

HSR-AIML-013 AI-Assisted 3D Commerce Scene Builder from Product Images and Text

Industry reference: Ctruh | 3D/AR SaaS, Developer Tools
Data: product images, captions and 3D assets from public repositories.
Technical work: vision-language tagging + asset retrieval + procedural scene composition.
Evaluation: Retrieval NDCG, scene completion time, user rating.
DIP direction: Image preprocessing/segmentation, feature or ViT/VLM embeddings, product-image retrieval/recognition.

HSR-AIML-014 WebAR Asset Optimization and Delivery Pipeline

Industry reference: Ctruh | 3D/AR SaaS, Developer Tools
Data: 3D models, textures, device/network profiles.
Technical work: mesh/texture optimization heuristics + adaptive delivery service.
Evaluation: Load time, FPS, bandwidth, visual quality score.
DIP direction: Image resizing/interpolation, texture quality, resolution/compression-quality trade-off, visual quality metrics.

HSR-AIML-015 Personalized Workout Recommendation and Adherence Prediction System

Industry reference: Cult.fit | HealthTech, Fitness, Consumer Tech
Data: workout history, goals, session metadata and feedback.
Technical work: hybrid recommendation + adherence/churn prediction.
Evaluation: NDCG@k, MAP@k, AUC, retention lift in simulation.

HSR-AIML-016 Computer-Vision Exercise Form Feedback Prototype

Industry reference: Cult.fit | HealthTech, Fitness, Consumer Tech
Data: pose-estimation datasets and recorded exercise videos.

Technical work: pose estimation + temporal classification + rule-based form scoring.
Evaluation: Keypoint error, action F1, form-error precision/recall.
DIP direction: Frame preprocessing, foreground/pose extraction, features/keypoints, CNN/pose model, temporal analysis.

HSR-AIML-017 IoT Water-Purifier Predictive Maintenance and Filter-Life Forecasting

Industry reference: DrinkPrime | IoT, Consumer Subscription, WaterTech
Data: sensor telemetry, service history, water-quality and usage signals.
Technical work: survival/RUL forecasting + anomaly detection.
Evaluation: MAE for RUL, recall for impending failures, false alarms.

HSR-AIML-018 Smart Service Dispatch Prioritization for Subscription IoT Fleets

Industry reference: DrinkPrime | IoT, Consumer Subscription, WaterTech
Data: device health alerts, technician locations, SLAs and service history.
Technical work: risk scoring + constrained scheduling/route optimization.
Evaluation: SLA breach reduction, travel distance, solve time.

HSR-AIML-019 Omnichannel Inventory Demand Forecasting and Stockout Prevention Engine

Industry reference: EasyEcom | E-commerce SaaS, Supply Chain
Data: SKU-store-channel sales, inventory and promotion histories.
Technical work: hierarchical forecasting + reorder recommendations.
Evaluation: WAPE/MAPE, stockout rate, inventory turns.

HSR-AIML-020 Real-Time Order Exception Detection Across E-commerce Channels

Industry reference: EasyEcom | E-commerce SaaS, Supply Chain
Data: order events, payment states, shipment updates and inventory events.
Technical work: Kafka-style stream processing + rule/ML anomaly detection.
Evaluation: Detection latency, precision/recall, throughput.

HSR-AIML-021 EV Charging Demand Forecasting and Station Congestion Prediction

Industry reference: ElectricPe | EV Mobility, Payments, Charging
Data: charging sessions, station metadata, time/weather/calendar features.
Technical work: demand forecasting + congestion probability model.
Evaluation: MAE/MAPE, Brier score, peak recall.

HSR-AIML-022 EV Charger Recommendation and Route-Aware Availability Service

Industry reference: ElectricPe | EV Mobility, Payments, Charging
Data: station locations, availability, connector types, user preferences and routes.
Technical work: ranking model + geospatial filtering + API service.
Evaluation: NDCG@k, availability hit-rate, API p95 latency.

HSR-AIML-023 B2B Account Intent Scoring and Next-Best-Account Ranking Engine

Industry reference: Factors.ai | MarTech, B2B Analytics, SaaS
Data: web events, CRM stages, campaign interactions and account metadata.
Technical work: learning-to-rank / propensity modelling.
Evaluation: NDCG@k, precision@k, lift over baseline.

HSR-AIML-024 Multi-Touch Marketing Attribution with Causal and Sequence Models

Industry reference: Factors.ai | MarTech, B2B Analytics, SaaS
Data: campaign touchpoints, conversions and spend.
Technical work: Markov/Shapley attribution plus sequence-model comparison.
Evaluation: Log loss, calibration, attribution stability, ROI consistency.

HSR-AIML-025 Real-Time UPI Transaction Risk Scoring for Youth-Focused Payments

Industry reference: Fam | FinTech, Payments, Consumer Tech
Data: synthetic payment events, device/session signals and merchant metadata.
Technical work: gradient boosting + graph/velocity features for fraud scoring.
Evaluation: PR-AUC, recall at low FPR, p95 latency.

HSR-AIML-026 Financial Habit Coach with Guardrailed Personalization

Industry reference: Fam | FinTech, Payments, Consumer Tech
Data: synthetic spend categories, budgets and educational content.
Technical work: rules + recommender + grounded LLM explanations; no personalized investment advice.
Evaluation: User-task success, hallucination rate, recommendation relevance.

HSR-AIML-027 Real-Time Customer 360 Identity Resolution and Event Unification Pipeline

Industry reference: FirstHive | MarTech, Customer Data Platform, AI
Data: multi-source customer events, CRM records and synthetic identities.
Technical work: deterministic/probabilistic entity resolution + Kafka/Spark pipeline.
Evaluation: Pairwise F1, cluster purity, throughput, end-to-end lag.
Relevant tools: Python; SQL; Spark; Kafka; Airflow/dbt; Snowflake/BigQuery; FastAPI; vector database; Docker/Kubernetes; AWS/GCP.

HSR-AIML-028 RAG-Based Customer Intelligence Copilot for CDP Users

Industry reference: FirstHive | MarTech, Customer Data Platform, AI
Data: customer schemas, campaign documentation, analytics definitions and synthetic account data.
Technical work: RAG + semantic search + agentic query planning over governed data.
Evaluation: Retrieval precision, groundedness, task completion, cost/latency.
Relevant tools: Python; SQL; Spark; Kafka; Airflow/dbt; Snowflake/BigQuery; FastAPI; vector database; Docker/Kubernetes; AWS/GCP.

HSR-AIML-029 Multimodal eKYC Identity Verification with Face, Document and Liveness Signals

Industry reference: FRS Labs | Identity, Fraud Detection, Cybersecurity
Data: public face/liveness/document datasets and synthetic identity records.
Technical work: face matching + OCR + liveness/deepfake checks + risk fusion.
Evaluation: FAR/FRR, ROC-AUC, document extraction F1.
DIP direction: Face/document enhancement, segmentation, feature extraction, CNN/ViT, image-quality and verification metrics.

HSR-AIML-030 Graph-Based Mule Account and Transaction Fraud Detection

Industry reference: FRS Labs | Identity, Fraud Detection, Cybersecurity
Data: synthetic banking transaction graph or public fraud graph datasets.
Technical work: graph features/GNN or boosted trees for suspicious network detection.
Evaluation: PR-AUC, recall@k, false-positive rate.

HSR-AIML-031 Intelligent Multi-Channel Notification Routing and Failover Engine

Industry reference: Fyno | Developer Tools, Communications Infrastructure
Data: synthetic delivery logs, provider latency/cost, user channel preferences.
Technical work: contextual ranking/bandit or rules for channel/provider selection.
Evaluation: Delivery success, p95 latency, cost/message.

HSR-AIML-032 Notification Infrastructure Observability and SLO Anomaly Detection

Industry reference: Fyno | Developer Tools, Communications Infrastructure
Data: API metrics, provider responses, queue depth and delivery receipts.
Technical work: streaming aggregation + anomaly detection + SLO burn-rate alerts.
Evaluation: MTTD, alert precision, throughput, availability.

HSR-AIML-033 SAR–Optical Multi-Sensor Change Detection for Disaster Intelligence

Industry reference: GalaxEye Space | SpaceTech, Geospatial AI, Remote Sensing

Data: sentinel-1 SAR + sentinel-2 optical imagery and public disaster labels.

Technical work: multi-branch fusion model for segmentation/change detection.

Evaluation: IoU, F1, precision/recall, calibration.

DIP direction: Multiband/FCC processing, registration/preprocessing, threshold/change maps, segmentation, CNN/ViT fusion.

Relevant tools: Python; PyTorch; OpenCV/rasterio; FastAPI; PostgreSQL/PostGIS; Docker; Kubernetes; GPU inference; AWS/on-prem.

HSR-AIML-034 Air-Gapped Geospatial ML Inference Platform for Multi-Sensor Imagery

Industry reference: GalaxEye Space | SpaceTech, Geospatial AI, Remote Sensing

Data: large raster tiles, trained CV models and synthetic analyst requests.

Technical work: containerized offline inference, tiling, batching, GPU scheduling, FastAPI services.

Evaluation: Throughput, p95 latency, GPU utilization, service availability.

DIP direction: Raster tiling, normalization, patch processing, segmentation/detection, stitching and accuracy/speed analysis.

Relevant tools: Python; PyTorch; OpenCV/rasterio; FastAPI; PostgreSQL/PostGIS; Docker; Kubernetes; GPU inference; AWS/on-prem.

HSR-AIML-035 EV Battery Remaining Useful Life Prediction from Telematics

Industry reference: Intellicar | Mobility, Fleet Telematics, EV Analytics

Data: public battery-cycling datasets plus synthetic fleet telemetry.

Technical work: RUL regression/survival model with uncertainty estimates.

Evaluation: MAE/RMSE, calibration, early-warning recall.

HSR-AIML-036 Fleet Driving Anomaly and Route Deviation Detection Platform

Industry reference: Intellicar | Mobility, Fleet Telematics, EV Analytics

Data: GPS traces, speed, ignition and trip telemetry.

Technical work: trajectory clustering + anomaly detection + geofencing.

Evaluation: Precision/recall, detection delay, false alerts per trip.

HSR-AIML-037 Personal Finance Transaction Categorization and Cash-Flow Forecasting Engine

Industry reference: Jupiter | FinTech, Neobanking

Data: synthetic/anonymous bank transaction descriptions and balances.

Technical work: text classification + cash-flow forecasting.

Evaluation: Macro-F1, MAE/MAPE, calibration.

HSR-AIML-038 Real-Time Payment Anomaly Detection with Explainable Risk Scores

Industry reference: Jupiter | FinTech, Neobanking

Data: synthetic transactions, device/session and merchant signals.

Technical work: streaming feature pipeline + boosted model + SHAP explanations.

Evaluation: PR-AUC, recall@FPR, p95 inference latency.

HSR-AIML-039 Offline-First Merchant Ledger Sync and Conflict Resolution Service

Industry reference: Khatabook | FinTech, SMB Software

Data: synthetic ledger edits from intermittently connected mobile clients.

Technical work: event sourcing/CRDT-inspired sync, idempotent APIs and conflict policies.

Evaluation: Sync correctness, conflict rate, p95 API latency.

Relevant tools: Java/Kotlin/Python/TypeScript; REST APIs; PostgreSQL; Docker; Playwright/Cypress/REST Assured; CI/CD.

HSR-AIML-040 AI-Assisted Test Automation Platform for Merchant Payment APIs

Industry reference: Khatabook | FinTech, SMB Software

Data: API specs, test fixtures, mock services and transaction workflows.

Technical work: LLM-assisted test generation + deterministic unit/integration/contract/E2E execution.

Evaluation: Coverage, defect detection rate, flaky-test rate, CI time.

Relevant tools: Java/Kotlin/Python/TypeScript; REST APIs; PostgreSQL; Docker; Playwright/Cypress/REST Assured; CI/CD.

HSR-AIML-041 Study-Abroad Program Matching and Explainable University Ranking

Industry reference: Leap Scholar | EdTech, FinTech, Study Abroad

Data: public university/course data, student preference profiles and constraints.

Technical work: learning-to-rank / hybrid recommender with transparent factor explanations.

Evaluation: NDCG@k, constraint satisfaction, diversity.

HSR-AIML-042 Document Intelligence for Application Readiness and Missing-Requirement Detection

Industry reference: Leap Scholar | EdTech, FinTech, Study Abroad

Data: synthetic applications, SOPs, transcripts and university requirement pages.

Technical work: OCR/NER + RAG checklist generator with evidence links.

Evaluation: Extraction F1, requirement recall, groundedness.

DIP direction: Document image preprocessing, contrast/thresholding, morphology, OCR and field extraction.

HSR-AIML-043 Restaurant Delivery Reconciliation and Revenue Leakage Detection Engine

Industry reference: Loop | Restaurant SaaS, Analytics, Reconciliation

Data: orders, aggregator settlements, refunds, commissions and payouts.

Technical work: entity matching + reconciliation rules + anomaly detection.

Evaluation: Match accuracy, leakage precision/recall, processing throughput.

HSR-AIML-044 Restaurant Channel Demand Forecasting and Promotion Impact Analytics

Industry reference: Loop | Restaurant SaaS, Analytics, Reconciliation

Data: order history, channel, menu, promotions and calendar data.

Technical work: forecasting + uplift/causal analysis.

Evaluation: WAPE/MAPE, uplift error, forecast bias.

HSR-AIML-045 AI Consumer Insight Mining from Reviews, Social Text and Survey Responses

Industry reference: Machine Vantage | AI, Consumer Science, CPG Analytics

Data: public product reviews, survey responses and social-text datasets.

Technical work: topic modelling/embeddings + aspect sentiment + insight clustering.

Evaluation: Topic coherence, sentiment F1, cluster quality.

HSR-AIML-046 Multimodal Concept Testing for CPG Product Innovation

Industry reference: Machine Vantage | AI, Consumer Science, CPG Analytics

Data: product images, packaging text, reviews and synthetic concept ratings.

Technical work: vision-language embeddings + preference prediction.

Evaluation: RMSE/AUC, retrieval NDCG, interpretability.

DIP direction: Colour/texture/shape preprocessing, image features, CNN/ViT/VLM embeddings, visual similarity/retrieval.

HSR-AIML-047 Enterprise Sensitive-Data Lineage and Exposure Risk Graph

Industry reference: Matters.ai | AI Data Security, Cybersecurity

Data: synthetic schemas, lineage metadata, access roles and data classifications.

Technical work: graph-based exposure scoring and attack-path analysis.

Evaluation: Risk ranking precision, path accuracy, update latency.

HSR-AIML-048 LLM Guardrail for Preventing Sensitive-Data Leakage in Enterprise Assistants

Industry reference: Matters.ai | AI Data Security, Cybersecurity

Data: prompt/response pairs, PII corpora, policy rules and synthetic secrets.

Technical work: PII/secret detection + policy engine + safe redaction before/after LLM.

Evaluation: Leakage recall, false-block rate, latency.

HSR-AIML-049 Camera-Only Road Scene Understanding for Indian Driving Conditions

Industry reference: Minus Zero | Autonomous Driving, Embodied AI
Data: BDD100K/KITTI/nuScenes plus india-relevant open driving imagery where licensed.
Technical work: multi-task detection/segmentation/lane perception model.
Evaluation: mAP, mIoU, lane F1, robustness by weather/light.
DIP direction: Edges/Hough vs learnt lane detection, object detection, semantic segmentation, CNN/ViT comparison.

HSR-AIML-050 Continual Learning Benchmark for Autonomous Driving Domain Shift

Industry reference: Minus Zero | Autonomous Driving, Embodied AI
Data: driving datasets partitioned by city/weather/time/domain.
Technical work: continual learning with replay/regularization and forgetting analysis.
Evaluation: Average accuracy, backward transfer, forgetting, calibration.
DIP direction: Brightness/contrast/weather normalization, augmentation, image-quality/domain analysis, CNN/ViT robustness.

HSR-AIML-051 Employee Shuttle ETA Prediction and Dynamic Route Optimization

Industry reference: MoveInSync | Mobility SaaS, Enterprise Transportation
Data: public/synthetic GPS traces, stops, employee pickup demand and traffic proxies.
Technical work: ETA regression + vehicle routing optimization.
Evaluation: MAE for ETA, on-time %, route distance, solver runtime.

HSR-AIML-052 Real-Time Fleet Safety and Route-Deviation Monitoring Platform

Industry reference: MoveInSync | Mobility SaaS, Enterprise Transportation
Data: GPS/speed/network telemetry and planned routes.
Technical work: streaming geospatial rules + clustering/anomaly detection.
Evaluation: Detection precision/recall, end-to-end latency, uptime.

HSR-AIML-053 Visitor Access Risk Scoring and Anomaly Detection for Gated Communities

Industry reference: MyGate | PropTech, Community Security, Consumer SaaS
Data: synthetic visitor entries, device/session signals and access outcomes.
Technical work: risk classification + anomaly detection with privacy-preserving features.
Evaluation: PR-AUC, false-positive rate, alert latency.

HSR-AIML-054 Resident Service Ticket Intelligence and SLA Prediction Platform

Industry reference: MyGate | PropTech, Community Security, Consumer SaaS
Data: synthetic community complaints, categories, timestamps and resolutions.
Technical work: text classification + priority/SLA regression + dashboard.
Evaluation: Macro-F1, MAE, SLA breach recall.

HSR-AIML-055 Generative Fashion Product Image Studio with Identity and Garment Consistency Checks

Industry reference: Neuropixel.ai | Generative AI, E-commerce Imaging
Data: open fashion datasets, garment images, poses and captions.
Technical work: conditional image generation/inpainting + similarity/consistency evaluation.
Evaluation: CLIP similarity, identity consistency, human preference.
DIP direction: Segmentation/masks, resizing/colour consistency, inpainting, perceptual/similarity evaluation.

HSR-AIML-056 Automated E-commerce Image Quality and Compliance Inspector

Industry reference: Neuropixel.ai | Generative AI, E-commerce Imaging
Data: product images with synthetic defects and marketplace-style requirements.
Technical work: image quality assessment + background/object/layout compliance classifiers.
Evaluation: Macro-F1, defect recall, false reject rate.
DIP direction: Contrast/sharpness/noise measures, segmentation, morphology, object/background analysis, quality classification.

HSR-AIML-057 Creative Performance Prediction and Automated Variant Ranking for Ads

Industry reference: Rocketium | AI Creative Automation, MarTech SaaS

Data: synthetic ad creatives, metadata and impression/click/conversion outcomes.

Technical work: multimodal embeddings + CTR/CVR prediction + ranking.

Evaluation: AUC, log loss, NDCG@k.

DIP direction: Image feature/embedding extraction, colour/layout/quality analysis, CNN/ViT/VLM features linked to prediction.

HSR-AIML-058 LLM-Agent Workflow for Brand-Safe Creative Adaptation Across Channels

Industry reference: Rocketium | AI Creative Automation, MarTech SaaS

Data: brand guidelines, campaign briefs, channel specs and creative assets.

Technical work: RAG + tool-using agent to generate/adapt copy with policy validation.

Evaluation: Groundedness, policy pass rate, task success, cost.

HSR-AIML-059 Contract Clause Extraction, Risk Classification and Evidence-Grounded Review Assistant

Industry reference: SpotDraft | LegalTech, AI SaaS

Data: public contract datasets, clause labels and legal playbooks.

Technical work: clause NER/classification + semantic retrieval + grounded LLM review.

Evaluation: Clause F1, retrieval precision, groundedness, hallucination rate.

Relevant tools: Python; TypeScript; FastAPI/Django; React/Next.js; PostgreSQL; Docker; vector database; GitHub Actions.

HSR-AIML-060 Contract Obligation Tracking and Workflow Automation Platform

Industry reference: SpotDraft | LegalTech, AI SaaS

Data: public contracts, synthetic obligations, owners and deadlines.

Technical work: obligation extraction + task workflow backend + TypeScript frontend.

Evaluation: Extraction F1, task completion, API p95 latency, test coverage.

Relevant tools: Python; TypeScript; FastAPI/Django; React/Next.js; PostgreSQL; Docker; vector database; GitHub Actions.

HSR-AIML-061 GenAI-Assisted Enterprise Support Portal with Java REST Services

Industry reference: Muraai Information Technologies | Enterprise Software, GenAI, Integration

Data: synthetic support tickets, FAQs and product documents.

Technical work: RAG assistant + classification routed through Java REST backend.

Evaluation: Macro-F1, groundedness, API latency.

Relevant tools: Java/J2EE; Python; SQL; REST/SOAP APIs; HTML/CSS; Docker; Git/GitHub; basic GenAI/RAG.

HSR-AIML-062 API Integration Testing and Data Reconciliation Suite for Enterprise Systems

Industry reference: Muraai Information Technologies | Enterprise Software, GenAI, Integration

Data: mock REST/SOAP APIs, SQL records and integration test cases.

Technical work: Java test harness + schema validation + reconciliation engine.

Evaluation: Test coverage, defect detection, reconciliation accuracy.

Relevant tools: Java/J2EE; Python; SQL; REST/SOAP APIs; HTML/CSS; Docker; Git/GitHub; basic GenAI/RAG.